

Notes: Sequeira, Nunn & Qian (RES, 2020)

Immigrants and the Making of America

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1 Big picture

- **Question.** What are the long-run economic effects of immigration during the U.S. Age of Mass Migration, 1850–1920?
- **Core empirical challenge.** Immigrants did not randomly choose counties; OLS can be biased upward or downward.
- **Instrument.** The paper uses the interaction between county railway access and national immigration booms.
- **Main result.** Counties with more historical immigration have higher income, less poverty, less unemployment, higher urbanization, and greater educational attainment in 2000.
- **Mechanisms.** The results point to manufacturing development, agricultural productivity, innovation, and eventual human-capital accumulation.
- **Economic takeaway.** Immigration generated local development effects that persisted for more than a century.

2 Data and measurement

2.1 Historical immigration, railways, and outcomes

County-level immigrant shares are measured from the U.S. Census for 1860–1920. The main immigration variable is the average foreign-born share across the census decades. Railway access is coded decade by decade using historical railroad maps. Contemporary outcomes include income, poverty, unemployment, urbanization, schooling, social capital, voting turnout, and crime.

Interpretation. The average historical immigrant share captures cumulative exposure to immigration during the Age of Mass Migration rather than a short-run one-year shock.

2.2 Instrument variables and controls

The main instrument is predicted average immigrant share generated by the interaction between lagged rail access and lagged aggregate immigrant inflows. The baseline controls include state fixed effects, cubic polynomials in latitude and longitude, date of railway connection, and predicted immigration based on industrialization and business cycles.

Interpretation. The controls are designed to remove general railroad, geography, industrialization, and business-cycle effects, leaving immigration-specific variation.

3 Models and empirical specifications

3.1 Zero-stage panel regression

$$ImmShare_{it} = \alpha_i + \delta_t + \beta (RailAccess_{i,t-1} \times ImmFlow_{t-1}) + \Gamma X_{it} + \varepsilon_{it}.$$

Interpretation. The zero-stage tests whether rail-connected counties received more immigrants specifically in decades when national immigrant inflows were high.

3.2 Predicted historical immigrant share

$$\widehat{ImmShare}_{it} = \hat{\beta} (RailAccess_{i,t-1} \times ImmFlow_{t-1}), \quad PredAvgImmShare_i = \frac{1}{T} \sum_t \widehat{ImmShare}_{it}.$$

Interpretation. This predicted average immigrant share is the IV. It uses only the interaction-driven part of immigrant settlement.

3.3 Cross-sectional 2SLS model

$$AvgImmShare_i = \pi PredAvgImmShare_i + \Theta Z_i + u_i, \\ Y_i = \rho AvgImmShare_i + \Theta Z_i + \eta_i.$$

Interpretation. The second-stage coefficient estimates the long-run effect of historical immigrant share on county outcomes.

4 Identification and empirical design

4.1 Core identification idea

Counties connected to railroads during immigration booms are compared with counties connected during immigration lulls. The identifying variation is not simply rail access and not simply aggregate immigration; it is their interaction.

Interpretation. The paper exploits timing. Being connected to the railroad matters for immigrant settlement only when many immigrants are arriving.

4.2 Validity checks

Table 1 checks balance between boom-connection and lull-connection counties. Table 2 validates the zero-stage. Table 3 reports the first stage. Tables 9 and 10 use a weather-shock instrument for aggregate immigrant inflows. Table 11 reports extensive robustness checks.

Interpretation. The design is credible because the same economic pattern appears across the baseline instrument, an alternative weather-based instrument, and many robustness restrictions.

4.3 Main threats

The main threats are railroad selection, business-cycle confounds, industrialization confounds, endogenous aggregate immigrant inflows, geographic confounds, and regional differences in the relevance of rail access. The paper responds with controls, fixed effects, region restrictions, alternative instruments, and robustness checks.

Interpretation. The estimates are best interpreted as local average effects for counties whose immigrant settlement was shifted by the timing of rail access during national immigration booms.

5 Tables and figures

5.1 Figure 1: Immigration into the U.S. during the Age of Mass Migration

Read: Panel (a) shows annual immigrant inflows; Panel (b) shows decadal inflows normalized by U.S. population. The boom/lull variation in Panel (b) is the national time-series component of the instrument.

Economic message. The figure visually motivates the instrument: national immigrant inflows fluctuate enough to generate differential exposure across railroad-connected counties.

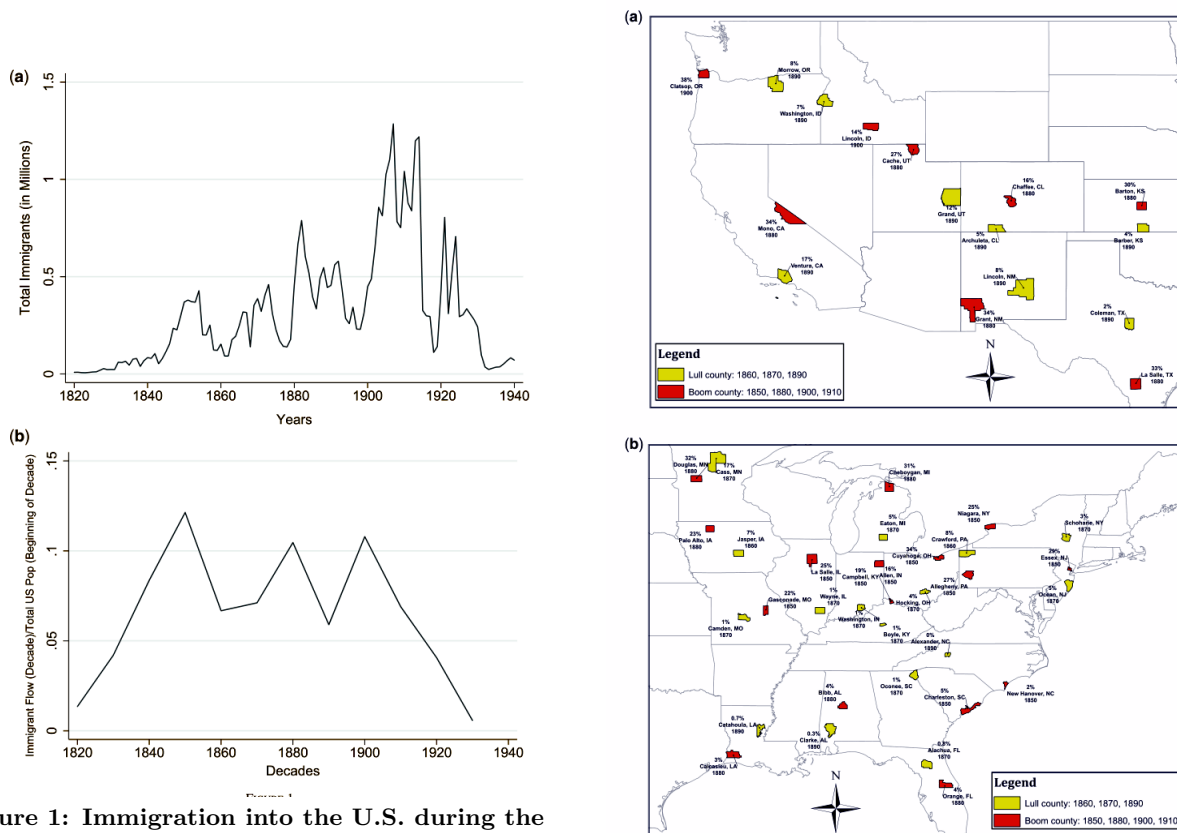


Figure 1: Immigration into the U.S. during the Age of Mass Migration. The caption is intentionally written as a reading guide: it states what the visual reports, what coefficient or pattern matters, and how the visual supports the paper's identification or mechanism.

Figure 2: Identification variation. The caption is intentionally written as a reading guide: it states what the visual reports, what coefficient or pattern matters, and how the visual supports the paper's identification or mechanism.

5.2 Figure 2: Illustration of the variation behind the identification strategy

Read: The maps pair counties within states, contrasting counties connected before immigration booms with counties connected before lulls.

Economic message. The figure makes clear that the design compares timing of railroad access relative to immigration booms, not rail access alone.

5.3 Table 1: Examining differences in baseline characteristics between lull- and boom-connection counties

Read: Most baseline demographic and economic characteristics are balanced; latitude and longitude differences motivate geographic controls.

Economic message. The table supports the plausibility of the quasi-experimental comparison.

TABLE 1
Examining differences in baseline characteristics between lull- and boom-connection counties

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
	Boom-connection counties			Lull-connection counties			Equality of means
Sample	Obs	Mean	Std Err	Obs	Mean	Std Err	<i>p</i> -value
Panel A: demographic composition							
Foreign share of the population, 1820	392	0.005	(0.0006)	312	0.004	(0.0005)	0.622
Foreign share of the population, 1830	524	0.004	(0.0005)	408	0.004	(0.0007)	0.482
Panel B: economic characteristics							
Urban share, 1840	626	0.975	(0.470)	496	0.695	(0.172)	0.575
Population density, 1830	670	0.147	(0.0480)	531	0.131	(0.022)	0.754
Share of the population in commerce, 1840	653	0.004	(0.0002)	509	0.005	(0.0003)	0.374
Share of the population in agriculture, 1840	653	0.259	(0.005)	509	0.260	(0.005)	0.916
Share of the population in mining, 1840	654	0.0009	(0.0002)	511	0.001	(0.0002)	0.582
Value of agricultural output per capita, 1840	663	46.332	(1.092)	527	44.25	(1.412)	0.244
Value of agricultural crops per capita, 1840	663	42.300	(1.076)	527	40.35	(1.404)	0.272
Post Offices per 1,000 inhabitants, 1840	672	0.698	(0.022)	536	0.652	(0.050)	0.403
Newspapers per 1,000 inhabitants 1840	242	0.175	(0.020)	120	0.112	(0.024)	0.048
Water connection indicator, 1840	670	0.467	(0.019)	531	0.514	(0.022)	0.106
Panel C: geographic characteristics							
Latitude	1,305	38.115	(0.125)	1,502	38.47	(0.124)	0.045
Longitude	1,305	-90.029	(0.306)	1,502	-92.164	(0.281)	0.000
Share of counties in the midwest and west	1,305	0.474	(0.014)	1,503	0.476	(0.013)	0.942
Share of counties in the south	1,305	0.436	(0.014)	1,503	0.460	(0.013)	0.195

Notes: "Boom-connection counties" are counties that we observe as connected to the railway for the first time in either 1850, 1880, or 1900. "Lull-connection counties" are counties that we observe as being connected to the railway for the first time in 1860, 1870, 1890, or 1910. Column 7 reports the *p*-value from a test of equality of means with unequal variances.

Table 1: Baseline balance. The caption is intentionally written as a reading guide: it states what the visual reports, what coefficient or pattern matters, and how the visual supports the paper's identification or mechanism.

5.4 Table 2 and Figure 3: Zero-stage evidence

Read: Table 2 shows that lag rail access interacted with lag immigrant inflows strongly predicts immigrant settlement. Figure 3 shows the bivariate version of the same relationship.

Economic message. These visuals validate the instrument's mechanism.

5.5 Table 3: First-stage relationship between predicted and actual immigrant shares

Read: The baseline first-stage coefficient is large and significant, with a Kleibergen–Paap statistic around 21.

Economic message. The predicted share is a strong instrument in the full sample.

5.6 Table 4: Effects of historical immigration on economic prosperity today

Read: The 2SLS estimates show higher income, lower poverty, lower unemployment, higher urbanization, and more schooling.

Economic message. This is the paper's main result: historical immigration raised long-run local prosperity.

Dependent variable	(1)	(2)	(3)	(4)	(5)	(6)
	All Counties	Excluding Northeast	Excluding South	Excluding South	South	Midwest and West
Interaction of interest:						
Lag Rail Access	0.172***	0.183***	0.046	0.239***	0.051*	0.292***
x Lag Immigrant Inflow/ Total US Pop	[0.045]	[0.051]	[0.107]	[0.076]	[0.027]	[0.085]
Other variables:						
Lag Rail Access	-0.002	0.010	0.087***	-0.056	-0.007	-0.038
	[0.015]	[0.015]	[0.015]	[0.036]	[0.009]	[0.040]
Lag Rail Access	-0.005	-0.009	-0.042***	0.014	0.002	0.007
	[0.006]	[0.006]	[0.006]	[0.014]	[0.004]	[0.016]
x Lag Log Industrialization Index	-0.001	-0.008	0.022	-0.023	-0.010	-0.028
	[0.010]	[0.010]	[0.027]	[0.027]	[0.008]	[0.027]
x Lag GDP Per Cap Decadal Growth	Yes	Yes	Yes	Yes	Yes	Yes
Lag Immigrant Share	Yes	Yes	Yes	Yes	Yes	Yes
Lag Urban Indicator	Yes	Yes	Yes	Yes	Yes	Yes
x Lag Immigrant Inflow/ Total US Pop	Yes	Yes	Yes	Yes	Yes	Yes
Log County Population Density	Yes	Yes	Yes	Yes	Yes	Yes
County Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes
Decade Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes
Observations	16,386	14,903	1,483	8,612	7,774	7,129
R-squared (within)	0.408	0.408	0.652	0.463	0.414	0.495
Mean Dependent Variable	0.085	0.080	0.138	0.144	0.021	0.145
SD Dependent Variable	0.109	0.109	0.088	0.112	0.057	0.116

Notes: OLS estimates are reported. An observation is a county in a time period (1860, 1870, 1880, 1890, 1900, 1910 or 1920). The dependent variable "Immigrant Share of Total County Population" is the proportion of a county's population that is foreign born in period t . "Lag Rail Access" is an indicator variable that equals one if a county has a railway in period $t-1$. Coefficients are reported, with Conley standard errors in square brackets. ***, **, and * indicate significance at the 1%, 5%, and 10% levels.

Table 2: Zero-stage OLS panel estimates. The caption is intentionally written as a reading guide: it states what the visual reports, what coefficient or pattern matters, and how the visual supports the paper's identification or mechanism.

Dependent variable	(1)	(2)	(3)	(4)	(5)	(6)
	All Counties	Excluding northeast	Excluding northeast	Excluding south	South	Midwest and west
Predicted Average Immigrant Share, 1860-1920	4.559***	4.184***	-3.585	9.730***	1.303**	8.903***
	[1.311]	[1.270]	[9.271]	[1.807]	[0.615]	[1.891]
Controls:						
Industrialization-Based Predicted Immig. Share	Yes	Yes	Yes	Yes	Yes	Yes
Business Cycle-Based Predicted Immig. Share	Yes	Yes	Yes	Yes	Yes	Yes
Date of RR Connection (Log Years as of 2000)	Yes	Yes	Yes	Yes	Yes	Yes
Cubic Polynomial for Latitude and Longitude	Yes	Yes	Yes	Yes	Yes	Yes
State Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes
Observations	2,935	2,720	215	1,593	1,342	1,378
Kleibergen Paap F -statistic	21.22	23.98	0.07	36.07	4.29	29.87
Standardized "beta" coeff. for Pred. Avg. Immig. Share	0.167	0.153	-0.095	0.311	0.099	0.285
Mean of Dependent Variable	0.098	0.095	0.137	0.160	0.024	0.163
SD of Dependent Variable	0.111	0.112	0.082	0.107	0.057	0.110

Notes: An observation is a county. Coefficient estimates are reported, with Conley standard errors reported in square brackets. ***, **, and * indicate significance at the 1%, 5% and 10% levels.

in the predicted immigrant share is associated with a 4.5 percentage point increase in the actual

Table 3: First stage. The caption is intentionally written as a reading guide: it states what the visual reports, what coefficient or pattern matters, and how the visual supports the paper's identification or mechanism.

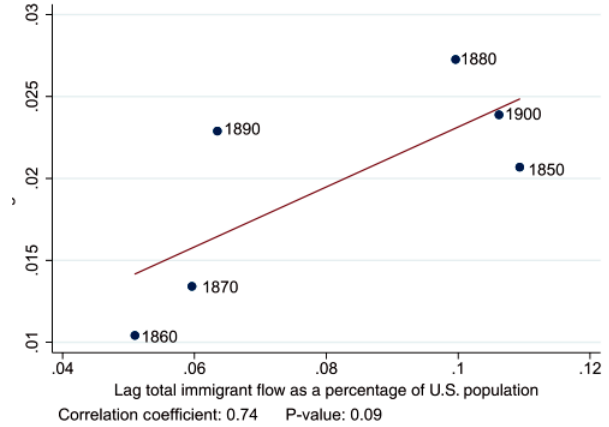


FIGURE 3

between the estimated effect of a county's connection to the railway on the foreign-born in a county and total immigration into the U.S.

Figure 3: Zero-stage bivariate relationship. The caption is intentionally written as a reading guide: it states what the visual reports, what coefficient or pattern matters, and how the visual supports the paper's identification or mechanism.

Dependent variable	(1)	(2)	(3)	(4)	(5)
	Log average per capita income, 2000	Share of Pop. below poverty line, 2000	Unemployment rate, 2000	Urbanization rate, 2000	Average years of schooling, 2000
A. OLS estimates					
Average Immigrant Share, 1860-1920	0.243*	0.015	0.020	0.949***	0.020
	[0.130]	[0.028]	[0.015]	[0.184]	[0.307]
B. Reduced form					
Predicted Average Immigrant Share, 1860-1920	11.942***	-2.229***	-1.876***	22.382***	41.925***
	[3.629]	[0.777]	[0.500]	[6.820]	[10.562]
C. 2SLS estimates					
Average Immigrant Share, 1860-1920	2.619***	-0.489**	-0.411***	4.909***	9.195***
	[1.022]	[0.209]	[0.151]	[2.008]	[3.392]
D. First stage estimates					
Dependent variable: average immigrant share, 1860-1920					
Predicted Average Immigrant Share, 1860-1920	4.559***	4.559***	4.559***	4.559***	4.559***
	[1.311]	[1.311]	[1.311]	[1.311]	[1.311]
Kleibergen Paap F -statistic	21.222	21.222	21.222	21.222	21.222
Controls (in all Panels):					
Industrialization-Based Predicted Immig. Share	Yes	Yes	Yes	Yes	Yes
Business Cycle-Based Predicted Immig. Share	Yes	Yes	Yes	Yes	Yes
Date of RR Connection (Log Years as of 2000)	Yes	Yes	Yes	Yes	Yes
Cubic Polynomial for Latitude and Longitude	Yes	Yes	Yes	Yes	Yes
State Fixed Effects	Yes	Yes	Yes	Yes	Yes
Observations	2,935	2,935	2,935	2,935	2,935
Mean of Dep. Var. (OLS, Reduced Form, and 2SLS)	10.022	0.136	0.047	0.401	11.445
SD of Dep. Var. (OLS, Reduced Form, and 2SLS)	0.203	0.054	0.025	0.305	0.558

Notes: An observation is a county. Panel A reports OLS estimates, Panel B reports reduced-form estimates, Panel C reports 2SLS estimates, and Panel D reports first-stage estimates. Coefficient estimates are reported, with Conley standard errors in square brackets. ***, **, and * indicate significance at the 1%, 5% and 10% levels.

Table 4: Long-run economic prosperity. The caption is intentionally written as a reading guide: it states what the visual reports, what coefficient or pattern matters, and how the visual supports the paper's identification or mechanism.

5.7 Table 5: Effects of historical immigration on social outcomes

Read: The IV estimates for social capital, voting turnout, and crime are small and statistically insignificant.

Economic message. The economic gains are not accompanied by robust long-run negative social outcomes in the measured variables.

TABLE 5
OLS and 2SLS estimates of the effects of historical immigration on social outcomes

Dependent variable	(1) Social Capital, 2000	(2) Voting Turnout, 2000	(3) Total Crime Rate, 2000	(4) Crimes Against Persons, 2000	(5) Crimes Against Property, 2000
A. OLS estimates					
Average Immigrant Share, 1860–1920	−0.048 [0.030]	−0.071 [0.046]	0.008*** [0.002]	0.002*** [0.001]	0.004*** [0.001]
B. Reduced form					
Predicted Average Immigrant Share, 1860–1920	0.210 [0.958]	1.244 [1.662]	0.086 [0.070]	0.020 [0.013]	0.054 [0.053]
C. 2SLS estimates					
Average Immigrant Share, 1860–1920	0.046 [0.209]	0.271 [0.347]	0.019 [0.017]	0.004 [0.003]	0.012 [0.012]
D. First-stage estimates					
Dependent variable: average immigrant share, 1860–1920					
Predicted Average Immigrant Share, 1860–1920	4.588*** [1.329]	4.596*** [1.330]	4.559*** [1.311]	4.559*** [1.311]	4.559*** [1.311]
Kleibergen Paap <i>F</i> -statistic	21.206	21.712	21.222	21.222	21.222
Controls (in all Panels):					
Industrialization-Based Predicted Immig. Share	Yes	Yes	Yes	Yes	Yes
Business Cycle-Based Predicted Immig. Share	Yes	Yes	Yes	Yes	Yes
Date of RR Connection (Log Years as of 2000)	Yes	Yes	Yes	Yes	Yes
Cubic Polynomial for Latitude and Longitude	Yes	Yes	Yes	Yes	Yes
State Fixed Effects	Yes	Yes	Yes	Yes	Yes
Observations	2,934	2,925	2,935	2,935	2,935
Mean of Dep. Var. (OLS, Reduced Form, and 2SLS)	0.182	0.535	0.006	0.001	0.004
SD of Dep. Var. (OLS, Reduced Form, and 2SLS)	0.061	0.090	0.004	0.001	0.003

Notes: An observation is a county. Panel A reports OLS estimates, Panel B reports reduced-form estimates, Panel C reports 2SLS estimates, and Panel D reports first-stage estimates. Coefficient estimates are reported, with Conley standard errors

Table 5: Social outcomes. The caption is intentionally written as a reading guide: it states what the visual reports, what coefficient or pattern matters, and how the visual supports the paper’s identification or mechanism.

5.8 Table 6: Effects on manufacturing output

Read: The estimates show positive effects on manufacturing output and establishments, especially by 1930.

Economic message. Manufacturing growth is one mechanism for persistent prosperity.

5.9 Table 7: Effects on farming

Read: The farming estimates are more mixed, but farm value per farm is positive in 1930.

Economic message. The agricultural channel supports the broader productivity interpretation, though less strongly than manufacturing.

5.10 Table 8: Effects on human capital and innovation

Read: Short-run education effects are negative, but patents are strongly positive; long-run schooling turns positive in the main contemporary results.

TABLE 6
OLS and 2SLS estimates of the effects of historical immigration on manufacturing output

Dependent variable	(1) Log Average Manufacturing Output per Capita	(2) Log Average Manufacturing Output per Establishment	(3) Log Average Manufacturing Output per Establishment	(4) Log Average Manufacturing Output per Establishment	(5) Log Number of Establishments per 10,000 Inhabitants	(6) Log Number of Establishments per 10,000 Inhabitants
	1860–1920	1930	1860–1920	1930	1860–1920	1930
A. OLS estimates						
Average Immigrant Share, 1860–1920	3.476*** [0.631]	4.216*** [0.796]	3.301*** [0.537]	3.343*** [0.648]	0.319** [0.249]	0.783*** [0.248]
B. Reduced form						
Predicted Average Immigrant Share 1860–1920	40.765 [33.988]	74.736*** [26.368]	20.778 [29.227]	71.924*** [23.653]	32.710*** [6.462]	2.079 [6.765]
C. 2SLS estimates						
Average Immigrant Share, 1860–1920	9.014 [8.460]	16.197*** [7.343]	4.594 [6.838]	15.588*** [6.868]	7.253*** [2.389]	0.453 [1.467]
D. First-stage estimates						
Dependent variable: average immigrant share, 1860–1920						
Predicted Average Immigrant Share, 1860–1920	4.523*** [1.381]	4.614*** [1.466]	4.523*** [1.381]	4.614*** [1.466]	4.510*** [1.381]	4.590*** [1.463]
Kleibergen Paap <i>F</i> -statistic	16.584	17.729	16.584	17.729	16.376	17.456
Controls (in all Panels):						
Industrialization-Based Predicted Immig. Share	Yes	Yes	Yes	Yes	Yes	Yes
Business Cycle-Based Predicted Immig. Share	Yes	Yes	Yes	Yes	Yes	Yes
Date of RR Connection (Log Years as of 2000)	Yes	Yes	Yes	Yes	Yes	Yes
Cubic Polynomial for Latitude and Longitude	Yes	Yes	Yes	Yes	Yes	Yes
State Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes
Observations	2,805	2,463	2,805	2,463	2,804	2,462
Mean of Dep. Var. (OLS, Reduced Form, and 2SLS)	6.561	7.206	12.578	14.030	3.352	2.487
SD of Dep. Var. (OLS, Reduced Form, and 2SLS)	1.197	1.419	0.906	1.156	0.536	0.509

Notes: An observation is a county. The decade of 1910 is missing from the Manufacturing Census. Panel A reports OLS estimates, Panel B reports reduced-form estimates, Panel C reports 2SLS estimates, and Panel D reports first-stage

Table 6: Manufacturing. The caption is intentionally written as a reading guide: it states what the visual reports, what coefficient or pattern matters, and how the visual supports the paper’s identification or mechanism.

TABLE 7
OLS and 2SLS estimates of the effect of historical immigration on farming

Dependent variable	(1)	(2)	(3)	(4)
	Log Average Total Farm Value (per Farm)		Log Average Total Farm Value (per Acre)	
	1860-1920	1930	1860-1920	1930
A. OLS estimates				
Average Immigrant Share, 1860-1920	0.571 [0.417]	1.321*** [0.340]	1.866*** [0.699]	2.224*** [0.721]
B. Reduced form				
Predicted Average Immigrant Share, 1860-1920	-1.771 [18.545]	32.991*** [13.341]	-12.372 [22.477]	14.961 [19.545]
C. 2SLS estimates				
Average Immigrant Share, 1860-1920	-0.393 [4.116]	7.455*** [3.485]	-2.743 [5.005]	3.367 [4.519]
D. First-stage estimates				
Dependent variable: average immigrant share, 1860-1920				
Predicted Average Immigrant Share, 1860-1920	4.510*** [1.381]	4.425*** [1.360]	4.510*** [1.381]	4.443*** [1.359]
Kleibergen Paap <i>F</i> -statistic	16.376	15.543	16.376	16.065
Controls (in all Panels):				
Industrialization-Based Predicted Immig. Share	Yes	Yes	Yes	Yes
Business Cycle-Based Predicted Immig. Share	Yes	Yes	Yes	Yes
Date of RR Connection (Log Years as of 2000)	Yes	Yes	Yes	Yes
Cubic Polynomial for Latitude and Longitude	Yes	Yes	Yes	Yes
State Fixed Effects	Yes	Yes	Yes	Yes
Observations	2,804	2,800	2,804	2,799
Mean of Dep. Var. (OLS, Reduced Form, and 2SLS)	10.420	11.513	5.907	6.558
SD of Dep. Var. (OLS, Reduced Form, and 2SLS)	0.913	0.829	0.701	0.793

Notes: An observation is a county. Panel A reports OLS estimates, Panel B reports reduced-form estimates, Panel C reports 2SLS estimates, and Panel D reports first-stage estimates. Coefficient estimates are reported, with Conley standard errors

TABLE 7
OLS and 2SLS estimates of the effect of historical immigration on farming

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	Log Average Total Farm Value (per Farm)		Log Average Total Farm Value (per Acre)	
	1860-1920	1930	1860-1920	1930
A. OLS estimates				
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B. Reduced form				
Predicted Average Immigrant Share, 1860-1920	-1.771 [18.545]	32.991*** [13.341]	-12.372 [22.477]	14.961 [19.545]
C. 2SLS estimates				
Average Immigrant Share, 1860-1920	-0.393 [4.116]	7.455*** [3.485]	-2.743 [5.005]	3.367 [4.519]
D. First-stage estimates				
Dependent variable: average immigrant share, 1860-1920				
Predicted Average Immigrant Share, 1860-1920	4.510*** [1.381]	4.425*** [1.360]	4.510*** [1.381]	4.443*** [1.359]
Kleibergen Paap <i>F</i> -statistic	16.376	15.543	16.376	16.065
Controls (in all Panels):				
Industrialization-Based Predicted Immig. Share	Yes	Yes	Yes	Yes
Business Cycle-Based Predicted Immig. Share	Yes	Yes	Yes	Yes
Date of RR Connection (Log Years as of 2000)	Yes	Yes	Yes	Yes
Cubic Polynomial for Latitude and Longitude	Yes	Yes	Yes	Yes
State Fixed Effects	Yes	Yes	Yes	Yes
Observations	2,804	2,800	2,804	2,799
Mean of Dep. Var. (OLS, Reduced Form, and 2SLS)	10.420	11.513	5.907	6.558
SD of Dep. Var. (OLS, Reduced Form, and 2SLS)	0.913	0.829	0.701	0.793

Notes: An observation is a county. Panel A reports OLS estimates, Panel B reports reduced-form estimates, Panel C reports 2SLS estimates, and Panel D reports first-stage estimates. Coefficient estimates are reported, with Conley standard errors

TABLE 8
OLS and 2SLS estimates of the effects of historical immigration on human capital and innovation

Dependent variable	(1)	(2)	(3)	(4)	(5)	(6)
	Educational attainment		Innovation		Log Patents per 10,000 Inhabitants:	
	Share Enrolled in School 1870-1920	Share Illiterate 1930	Share 1870-1920	Share 1930	All Nationalities 1860-1920	European 1860-1920
A. OLS estimates						
Average Immigrant Share, 1860-1920	-0.119*** [0.015]	-0.059*** [0.016]	0.103*** [0.035]	0.044*** [0.013]	1.379*** [0.474]	2.992*** [0.555]
B. Reduced form						
Predicted Average Immigrant Share, 1860-1920	-3.350*** [1.009]	-1.314*** [0.451]	7.558*** [3.190]	1.415*** [0.781]	139.378*** [21.909]	38.047*** [9.336]
C. 2SLS estimates						
Average Immigrant Share, 1860-1920	-0.735*** [0.308]	-0.288*** [0.113]	1.658** [0.862]	0.310** [0.123]	30.366*** [9.277]	8.289*** [1.998]
D. First-stage estimates						
Dependent variable: average immigrant share, 1860-1920						
Predicted Average Immigrant Share, 1860-1920	4.559*** [1.311]	4.559*** [1.311]	4.559*** [1.311]	4.559*** [1.311]	4.590*** [1.332]	4.590*** [1.332]
Kleibergen Paap <i>F</i> -statistic	21.222	21.222	21.222	21.222	21.151	21.151
Controls (in all Panels):						
Industrialization-Based Predicted Immig. Share	Yes	Yes	Yes	Yes	Yes	Yes
Business Cycle-Based Predicted Immig. Share	Yes	Yes	Yes	Yes	Yes	Yes
Date of RR Connection (Log Years as of 2000)	Yes	Yes	Yes	Yes	Yes	Yes
Cubic Polynomial for Latitude and Longitude	Yes	Yes	Yes	Yes	Yes	Yes
State Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes
Observations	2,935	2,935	2,935	2,935	2,929	2,929
Mean of Dep. Var. (OLS, Reduced Form, and 2SLS)	0.190	0.217	0.104	0.041	3.561	0.312
SD of Dep. Var. (OLS, Reduced Form, and 2SLS)	0.035	0.026	0.104	0.042	1.263	0.589

Notes: An observation is a county. Panel A reports OLS estimates, Panel B reports reduced-form estimates, Panel C reports 2SLS estimates, and Panel D reports first-stage estimates. Coefficient estimates are reported, with Conley standard errors

Table 8: Human capital and innovation. The caption is intentionally written as a reading guide: it states what the visual reports, what coefficient or pattern matters, and how the visual supports the paper's identification or mechanism.

Economic message. Immigration stimulated innovation and eventually human-capital accumulation, despite lower short-run average education.

5.11 Tables 9 and 10: Weather-shock predicted immigrant inflows

TABLE 9
OLS and 2SLS estimates of the effects of historical immigration on economic prosperity today, using immigrant inflows predicted by sending-country weather shocks

Dependent variable	(1) Log Average per Capita Income, 2000	(2) Share of Pop. Below Poverty Line, 2000	(3) Unemployment Rate, 2000	(4) Urbanization Rate, 2000	(5) Average Years of Schooling, 2000
A. OLS estimates					
Average Immigrant Share, 1860–1920	0.244* [0.130]	0.015 [0.028]	0.020 [0.015]	0.948*** [0.185]	0.022 [0.307]
B. Reduced form					
Predicted Average Immigrant Share, 1860–1920	28.934*** [9.683]	−4.571 [2.874]	−4.402*** [1.453]	65.007*** [19.282]	115.574*** [27.643]
C. 2SLS estimates					
Average Immigrant Share, 1860–1920	2.792*** [1.249]	−0.441 [0.298]	−0.425*** [0.195]	6.273*** [2.651]	11.152*** [4.397]
D. First-stage estimates					
Dependent variable: average immigrant share, 1860–1920					
Predicted Average Immigrant Share, 1860–1920	10.364*** [3.058]	10.364*** [3.058]	10.364*** [3.058]	10.364*** [3.058]	10.364*** [3.058]
Kleibergen Paap F-statistic	18.006	18.006	18.006	18.006	18.006
Controls (in all Panels):					
Industrialization-Based Predicted Immig. Share	Yes	Yes	Yes	Yes	Yes
Business Cycle-Based Predicted Immig. Share	Yes	Yes	Yes	Yes	Yes
Date of RR Connection (Log Years as of 2000)	Yes	Yes	Yes	Yes	Yes
Cubic Polynomial for Latitude and Longitude	Yes	Yes	Yes	Yes	Yes
State Fixed Effects	Yes	Yes	Yes	Yes	Yes
Observations	2,935	2,935	2,935	2,935	2,935
Mean of Dep. Var. (OLS, Reduced Form, and 2SLS)	10.022	0.136	0.047	0.401	11.445
SD of Dep. Var. (OLS, Reduced Form, and 2SLS)	0.203	0.054	0.025	0.305	0.558

Notes: An observation is a county. Panel A reports OLS estimates, Panel B reports reduced-form estimates, Panel C reports 2SLS estimates, and Panel D reports the first-stage estimates. Coefficient estimates are reported, with Conley standard errors in *curly braces*. *** ** and * indicate significance at the 1%, 5% and 10% levels.

Table 9: Weather-shock instrument and prosperity. The caption is intentionally written as a reading guide: it states what the visual reports, what coefficient or pattern matters, and how the visual supports the paper’s identification or mechanism.

Read: The alternative instrument based on sending-country weather shocks yields similar economic and social outcome patterns.

Economic message. The results are not driven solely by U.S.-side determinants of aggregate immigrant inflows.

5.12 Table 11: Current income robustness checks

Read: The income effect remains positive under alternative instruments, controls, and restricted samples.

Economic message. The main income result is robust and not driven by a single specification or sample restriction.

TABLE 10
OLS and 2SLS estimates of the effects of historical immigration on social outcomes, using immigrant inflows predicted by sending-country weather shocks

Dependent variable	(1) Social Capital, 2000	(2) Voting Turnout, 2000	(3) Total Crime Rate, 2000	(4) Crimes Against Persons, 2000	(5) Crimes Against Property, 2000
A. OLS estimates					
Average Immigrant Share, 1860–1920	−0.048 [0.030]	−0.071 [0.046]	0.008*** [0.002]	0.002*** [0.0005]	0.004*** [0.001]
B. Reduced form					
Predicted Average Immigrant Share, 1860–1920	1.506 [2.838]	2.733 [5.032]	0.348 [0.215]	0.063 [0.039]	0.241 [0.162]
C. 2SLS estimates					
Average Immigrant Share, 1860–1920	0.144 [0.278]	0.254 [0.460]	0.034 [0.024]	0.006 [0.005]	0.023 [0.018]
D. First-stage estimates					
Dependent variable: average immigrant share, 1860–1920					
Predicted Average Immigrant Share, 1860–1920	10.424*** [3.097]	10.776*** [3.161]	10.364*** [3.058]	10.364*** [3.058]	10.364*** [3.058]
Kleibergen Paap F-statistic	17.806	18.143	18.006	18.006	18.006
Controls (in all Panels):					
Industrialization-Based Predicted Immig. Share	Yes	Yes	Yes	Yes	Yes
Business Cycle-Based Predicted Immig. Share	Yes	Yes	Yes	Yes	Yes
Date of RR Connection (Log Years as of 2000)	Yes	Yes	Yes	Yes	Yes
Cubic Polynomial for Latitude and Longitude	Yes	Yes	Yes	Yes	Yes
State Fixed Effects	Yes	Yes	Yes	Yes	Yes
Observations	2,934	2,925	2,935	2,935	2,935
Mean of Dep. Var. (OLS, Reduced Form, and 2SLS)	0.182	0.535	0.006	0.001	0.004
SD of Dep. Var. (OLS, Reduced Form, and 2SLS)	0.061	0.090	0.004	0.001	0.003

Notes: An observation is a county. Panel A reports OLS estimates, Panel B reports reduced-form estimates, Panel C reports 2SLS estimates, and Panel D reports first-stage estimates. Coefficient estimates are reported, with Conley standard errors in *curly braces*.

Table 10: Weather-shock instrument and social outcomes. The caption is intentionally written as a reading guide: it states what the visual reports, what coefficient or pattern matters, and how the visual supports the paper’s identification or mechanism.

Dependent variable	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
	Log Average per Capita Income, 2000								
	Variants of Instrument			Additional Covariates			Restricted Samples		
	Baseline Specification	Average in Connected Decades	Immigrant Inflow in First Decade	No Covariates	Frontier Experience	Civil War	Internet Trade	Removing Outliers	Counties w Constant Borders
A. OLS estimates									
Average Immigrant Share, 1860–1920	0.243* [0.130]	0.246* [0.133]	0.269** [0.130]	0.395*** [0.134]	0.220 [0.128]	0.244* [0.129]	0.225*** [0.087]	0.333*** [0.087]	0.607*** [0.193]
B. Reduced form									
Predicted Average Immigrant Share, 1860–1920	11.942*** [3.629]	287.404*** [17.292]	49.319*** [12.435]	12.719*** [2.293]	9.793*** [3.413]	12.183*** [3.618]	10.577*** [2.912]	14.684*** [3.773]	15.778*** [5.794]
C. 2SLS estimates									
Average Immigrant Share, 1860–1920	2.619*** [1.022]	8.532* [4.497]	4.540*** [1.771]	2.865*** [0.790]	2.238*** [0.964]	2.603*** [1.035]	3.564*** [1.581]	3.831*** [1.466]	3.119** [1.015]
D. First-stage estimates									
Predicted Average Immigrant Share, 1860–1920	4.559*** [1.311]	33.686* [17.511]	10.862*** [3.275]	4.439*** [1.214]	4.375*** [1.301]	4.524*** [1.315]	2.954*** [1.132]	3.833*** [1.324]	5.059*** [2.129]
Kleibergen-Paap F -statistic	21.222	18.803	21.085	75.973	19.863	21.256	21.225	17.027	10.675
Controls (in all Panels):									
Never Connected to the Railroad (0–1)	No	Yes	Yes	No	No	No	No	No	No
Number of Years with Frontier Experience	No	No	No	No	Yes	No	No	No	No
County Connected During Civil War (1860)	No	No	No	No	No	Yes	No	No	No
Trade-Based Predicted Immigrant Share	No	No	No	No	No	Yes	No	No	No
Industrialization-Based Predicted Immigrant Share	Yes	Yes	Yes	No	Yes	Yes	No	Yes	Yes
Business Cycle-Based Predicted Immigrant Share	Yes	Yes	Yes	No	Yes	Yes	Yes	Yes	Yes
Date of RR Connection (Log Years as of 2000)	Yes	No	No	No	Yes	Yes	Yes	Yes	Yes
Cubic Polynomial for Latitude and Longitude	Yes	Yes	Yes	No	Yes	Yes	Yes	Yes	Yes
State Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	2,935	2,935	2,935	2,935	2,934	2,935	2,935	2,761	1,489
Mean of Dep. Var. (OLS, Reduced Form, and 2SLS)	10.022	10.022	10.022	10.022	10.022	10.022	10.022	10.013	10.021
SD of Dep. Var. (OLS, Reduced Form, and 2SLS)	0.203	0.203	0.203	0.203	0.203	0.203	0.203	0.170	0.205

Notes: An observation is a county. Panel A reports OLS estimates, Panel B reports reduced form estimates, Panel C reports 2SLS estimates, and Panel D reports the first-stage estimates from the 2SLS. Coefficient estimates are reported, with Cooley standard errors reported in square brackets. ***, **, and * indicate significance at the 1%, 5% and 10% levels.

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Table 11: Robustness checks. The caption is intentionally written as a reading guide: it states what the visual reports, what coefficient or pattern matters, and how the visual supports the paper’s identification or mechanism.

6 Short summary

- The paper studies long-run effects of the Age of Mass Migration.
- The IV uses rail access interacted with national immigration booms.
- Historical immigration raises current income, lowers poverty and unemployment, increases urbanization, and increases schooling.
- Social outcomes show no robust negative long-run effects.
- Mechanisms include manufacturing, agriculture, innovation, and human capital.
- Alternative weather-based instruments and robustness checks support the main interpretation.

7 Conclusion

7.1 Core conclusion

Historical immigration had persistent positive effects on local economic development in the U.S.

7.2 Economic intuition

Immigrants contributed to industrialization, innovation, urban growth, and later human-capital accumulation. These channels can generate persistent development paths through agglomeration and productivity spillovers.

7.3 Identification takeaway

The instrument is powerful because it combines local rail access with national immigration booms and lulls, isolating immigrant settlement variation that is not simply railroad access or county selection.

7.4 Limitations

The estimates are local to the historical U.S. setting and to counties affected by rail-timing variation. The mechanisms are suggestive rather than a complete decomposition.

7.5 Bottom line

The paper shows that immigration can have long-lasting local development benefits without robust evidence of persistent social costs in the measured outcomes.